

Translation with LLMs through Prompting with Long-Form Context (Part 3)

5. Test Case Example (Continued):

- **Proposed Prompt Input (step 3):**

Which of the following sentences are translations of the following English sentence? Multiple sentences can be chosen.
[English] This is an example.

1. Este es un enlace de ejemplo.
2. Cursiva y negrita: utilice asteriscos y guiones bajos para indicar intervalos de énfasis.
3. Utilice un asterisco (o guión bajo) para cursiva y dos asteriscos (o dos guiones bajos) para negrita.
4. Por ejemplo...

- **Proposed Prompt Expected Output:**

The sentence "This is an example." can be translated to Spanish as:

1. Este es un ejemplo.
2. Por ejemplo...

These two options correctly translate the meaning of "This is an example." into Spanish.

6. Fallback Plan: If the proposed method does not help as compared to the baseline, analyzing the results of step 3 would likely provide further insights into how the template should be modified. In addition to potentially identifying off-target errors, it may be that the model is unable to identify correct translations even if they have been generated, and results are likely to vary across languages based on their training data. Using the generated paragraph as provided context and still querying the model to translate at only the sentence level could be compared. Restricting monolingual text to be retrieved text within the domain of the source sentence could be explored. Adding few-shot examples in the prompt and comparing other MT prompt templates may also help debug the proposed method. Including an additional query where the model is first asked to label each generated token by langid and then asked to re-translate the source including those tokens which are correctly labelled in target may reinforce langid and guide generation in the target language. Performing layer-wise analyses of likelihood of generating the next token in-language and in-script for open-weight models may also help debug where and why off-target issues persist.

Reviewer 1

Novelty: 5 (somewhat novel - there are differences from existing ideas but not enough to turn into a new paper)

Rationale: While I'm not aware of papers that have used this exact prompting strategy, I don't think that this proposal will be enough to justify a publication. I think that there should be a variety of strategies suggested + an analysis of multiple prompting strategies rather than suggesting one strategy. I think that a thorough analysis of the effects of additional context / langids could potentially turn this into a paper.

Feasibility: 9

Rationale: Such a project that only uses LLM APIs could be executed very quickly without much expertise in coding/architecture. The only time-consuming part might be iterating and adjusting the prompts in the ablation studies.

Expected Effectiveness: 7

Rationale: I think that this proposal could work well to guide LLMs to translate in the desired target language, since this is a known problem with current prompt-based MT strategies (as the writers have suggested).

Excitement: 5 (Leaning negative: it has interesting bits but overall not exciting enough)

Rationale: I'm not sure how well this method will transfer to future models, and this could be a limiting factor in the longevity of this research. (But this is a limitation of all prompting research...)

Overall Score: 5 (Decent idea but has some weaknesses or not exciting enough, marginally below the acceptance threshold of major AI conferences)

Rationale: I think that the work should focus on the ablation studies and comparison of multiple prompting strategies / analysis, rather than focusing on one new strategy.

Confidence: 3 (You are fairly confident that the evaluation is correct)

Reviewer 2

Novelty: 1 (not novel at all - there are many existing ideas that are the same)

Rationale: There are multiple existing works on prompting LLMs on low-resource translation, usually using few-shot demo. <https://proceedings.mlr.press/v202/garcia23a/garcia23a.pdf> <https://arxiv.org/pdf/2305.14857>
Also work explaining why few-shot prompt would work: <https://arxiv.org/pdf/2305.10266>

Feasibility: 5 (Moderately feasible: It can probably be executed within the given time frame but would require careful planning, efficient use of APIs or some advanced computational strategies to overcome the limited GPU resources, and would require some modifications to the original proposal to make it work.)

Rationale: The prompting experiment is mostly feasible given one can afford the API calls. The model, prompts, and evaluation metrics are concrete, although unclear if the proposed experiment is useful for proving the research idea, e.g., a few high-resource languages are listed for a research idea that focuses on low-resource languages.

Expected Effectiveness: 3 (Low Effectiveness: The idea might work in some special scenarios but you don't expect it to work in general.)

Rationale: The proposed experiment can help find a set of relatively high-performing prompts, but it is unclear among the prompts proposed if any of them will bring any improvement.

Excitement: 3 (Mediocre: this idea makes marginal contributions and is very incremental)

Rationale: The ability to do prompting/few-shot translation is fundamentally tied to the training data, see <https://arxiv.org/pdf/2305.10266>, so trying to solve this problem from the prompting space is inherently limited.

Overall Score: 3 (Clear rejection for major AI conferences)

Rationale: There is similar work on prompting LLMs to generate translation in low-resource languages, hence the idea is not very novel. Moreover, in terms of the goal to generate high-quality low-resource translation, the gains likely are not going to come from prompting.

Confidence: 4 (You are confident but not absolutely certain that the evaluation is correct)

S Example Idea: Linguistic Pivot Constellation: Enhancing Cross-Lingual Transfer for Low-Resource Languages and Dialects

Linguistic Pivot Constellation (LPC): Enhancing Cross-Lingual Transfer for Low-Resource Languages and Dialects (Part 1)

1. Problem Statement: Large language models struggle with cross-lingual transfer, especially for low-resource languages and dialects. This limitation hinders the models' ability to perform well on multilingual tasks involving these languages, potentially exacerbating digital language divides.

2. Motivation: Current approaches often rely on parallel data or multilingual pretraining, which are limited for many language pairs. Inspired by how polyglots leverage similarities between known languages to learn new ones, we propose creating a network of conceptual bridges across languages. This method could potentially overcome the limitations of existing approaches by leveraging the model's broad knowledge to create connections between known and unknown linguistic territories.

3. Proposed Method: We introduce Linguistic Pivot Constellation (LPC), a novel prompting technique that constructs a dynamic network of linguistic pivot points. For a given task, LPC first identifies conceptually similar languages or dialects to the target language. It then generates a constellation of prompts in these pivot languages, each capturing a different aspect of the task. The model is guided to 'triangulate' the correct response by considering these multiple perspectives. For example, to translate a rare dialect, LPC might use prompts in related languages, regional lingua francas, and even etymologically connected languages.

4. Step-by-Step Experiment Plan:

1. Data Collection

- Gather datasets for translation and question-answering tasks across a diverse set of low-resource languages and dialects.
- Utilize the FLORES-101 dataset for machine translation and the TyDi QA dataset for question answering.

2. Baseline Implementation

- Implement standard few-shot prompting and existing cross-lingual transfer methods (e.g., zero-shot cross-lingual transfer) as baselines.

3. LPC Implementation

- (a) Create a language similarity matrix based on language families and geographical proximity.
- (b) Implement a function to select the most relevant pivot languages for a given target language.
- (c) Design prompts for each pivot language that capture different aspects of the task.

4. Prompt Construction

- (a) Select 3-5 pivot languages based on the similarity matrix.
- (b) Generate task-specific prompts in each pivot language.
- (c) Combine these prompts into a 'constellation' prompt that includes the original task in the target language.

5. Model Selection

- Use GPT-4 as the primary model for experiments.
- Test with GPT-3.5-turbo for comparison.

6. Experiment Execution

- (a) Run the baseline methods.
- (b) Run the LPC method with varying numbers of pivot languages (1, 3, and 5).
- (c) Record the model outputs and performance metrics.

Linguistic Pivot Constellation (LPC): Enhancing Cross-Lingual Transfer for Low-Resource Languages and Dialects (Part 3)

4. Step-by-Step Experiment Plan (Continued):

7. Evaluation

- Evaluate the results using task-specific metrics:
 - BLEU score for translation tasks
 - F1 score for question answering tasks

8. Analysis

- Analyze the effectiveness of different pivot language combinations and the method's scalability to extremely low-resource scenarios.
- Compare LPC performance against baselines across different language families and resource levels.

5. Test Case Examples:

• Test Case 1:

- **Baseline Prompt Input:** Translate the following Sicilian sentence to English: 'Unni c'è fumu c'è focu.'
- **Baseline Prompt Expected Output:** Where there's smoke, there's fire.
- **Proposed Prompt Input:** We will translate a Sicilian sentence to English. To help with this task, consider the following related phrases:

In Italian: 'Dove c'è fumo c'è fuoco.'

In Neapolitan: 'Addò ce sta 'o fummo ce sta 'o ffuoco.'

In Latin: 'Ubi fumus, ibi ignis.'

Now, translate the Sicilian sentence to English: 'Unni c'è fumu c'è focu.'

- **Proposed Prompt Expected Output:** Where there's smoke, there's fire.
- **Explanation:** The LPC method provides context from related languages (Italian, Neapolitan, and Latin), which can help the model better understand and translate the Sicilian phrase. This is especially useful for low-resource languages like Sicilian, where direct translation data might be limited.

6. Fallback Plan: If the LPC method does not significantly outperform baselines, we will pivot the project towards an in-depth analysis of cross-lingual transfer mechanisms. We will investigate the relationship between language similarity and transfer effectiveness, the impact of pivot language selection on performance, and how different aspects of language (lexical, syntactic, semantic) transfer across the constellation. This analysis could provide valuable insights into the strengths and limitations of large language models in cross-lingual tasks, potentially informing future research directions in multilingual Natural Language Processing.